EMERSON EIMS
BUILDING SUITE PRO

Validation Engine Test

Location: Nairobi, Kenya
Total Area: 450 m² | Units: 3 | Storeys: 2
Architectural Style: Modern
Date: 26 April 2026

PROJECT INFORMATION	
Project ID	45729858-2593-4734-b669-edd044ac4ec0
Location	Nairobi, Kenya
Total Area	450 m²
Units / Apartments	3
Storeys	2
Bedrooms per Unit	3
Foundation Type	Strip + Ground Beam
Style	Modern
Report Generated	2026-04-26 21:34

Validation Engine — 3 Auto-Correction(s) Applied

EIMS does not silently template-fill. The validation engine reviewed every input, detected the issues below, applied a correction with code citation, and propagated it through all downstream calculations. The report you are reading is internally consistent.

Field	Original	Auto-corrected to	Reason
Location	Not specified	**Nairobi, Kenya**	Mandatory for county submission — defaulted to capital city.
Currency	USD	**KES**	Kenyan locality detected — CBK Cap.491 requires KES for stat
Foundation	strip	**strip + ground beam**	2-storey → strip + ground beam (BS 8004 §7)

Compliance Status: AUTO-CORRECTED

TABLE OF CONTENTS

1. PROJECT SUMMARY

Phase phase_1: Site Analysis

water_table_depth	3.5
water_table_source	regional_estimate
water_table_note	Regional estimate from climate/soil model -- verify with on-site geotechnical su
soil_type	clay
vegetation_coverage	0.65
slope_angle	8.25
suitability_pct	90
nearby_structures	survey_required
land_use	residential
climate_zone	tropical
analysis_method	regional_data_model
data_source	EIMS Regional Database + GPS Coordinates

Phase phase_2: Geotechnical Design

bearing_capacity_kPa	150
soil_classification	CH
recommended_foundation	STRIP
settlement_mm	40
slope_stability	stable
water_table_depth	3.5
excavation_depth	2.8

Phase phase_3: Foundation Design

foundation_type	PAD
foundation_depth_m	1.65
footing_width_m	2.03
total_building_load_kN	1650.0
bearing_capacity_kPa	150
safety_factor	3.0
concrete_volume_m3	54.45
reinforcement_steel_kg	4900.5
soil_type	clay
water_table_factor	standard
design_standard	BS 8004 / Eurocode 7
bim_ready	True

Phase phase_4: Floor Plans

floor_count	3
total_area_sqm	220.0
bathrooms	3
rooms	{'bedrooms': 4, 'bathrooms': 3, 'living_areas': 1, 'kitchen': 1, 'extra_rooms':

style	modern
area_per_unit	220.0
building_code_compliance	True
area_efficiency_percent	80.0
cost_per_sqm_estimate	calculated_in_costing_phase

Phase phase_5: Electrical Design

total_connected_load_kW	9.38
max_demand_kW	5.63
diversity_factor	0.6
supply_type	single-phase 230V
main_current_A	24.5
lighting_points	12
socket_outlets	12
total_circuits	7
lighting_circuits	2
socket_circuits	2
fixed_appliance_circuits	3
distribution_boards	1

Phase phase_6: Plumbing Design

total_occupants	6
daily_demand_liters	900
peak_hourly_liters	90
total_fixture_units	34.5
main_supply_pipe_mm	32
cold_water_pipes	11
hot_water_pipes	7
waste_outlets	10
soil_stacks	1
hot_water_storage_liters	240
water_heater_kW	9.6
storage_tank_liters	1000

Phase phase_7: BOQ

material_items	77
total_material_cost	60448.52
currency	USD
items_by_category	{'structural': 17, 'plumbing': 14, 'electrical': 11, 'mep_hvac': 5, 'finishes':
items	[{'item_no': 1, 'description': 'Portland Cement (50kg)', 'unit': 'bag', 'quantit
calculation_basis	{'total_area_m2': 220.0, 'stories': 3, 'units': 1, 'bedrooms_per_unit': 4, 'foot
waste_factor_applied	True
method	quantity_takeoff_from_dimensions

Phase phase_8: Infrastructure

solar_system	{'panels_kw': 4.4, 'panels_count': 11, 'battery_kWh': 6.6, 'estimated_cost_usd':
borehole_system	{'depth_meters': 8, 'estimated_yield_lpm': 15, 'pump_kw': 3, 'estimated_cost_usd
grid_option	{'connection_cost_usd': 260, 'monthly_tariff_estimate_usd': 33}
recommendation	solar_grid_hybrid
estimated_annual_saving_usd	819.84

Phase phase_9: Landscape Design

lawn_area_m2	77.0
hardscape_area_m2	22.0
garden_area_m2	33.0
plants_count	264
irrigation_system	True
recommended_style	tropical_lush
estimated_cost_usd	759
maintenance_monthly_usd	15

Phase phase_10: Permits & Compliance

jurisdiction	Kenya
building_code	Kenya Building Code 2020
building_code_compliance	REQUIRES_VERIFICATION
permits_required	6
estimated_timeline_days	45
estimated_fees_usd	528
setback_compliance	{'front_m': 5, 'side_m': 2, 'rear_m': 3}
required_documents	['Architectural drawings (stamped)', 'Structural calculations (engineer-certifie

Phase phase_11: 3D Visualization

status	ready
model_endpoint	/api/drawings/3d-model

Phase phase_12: Live Costing

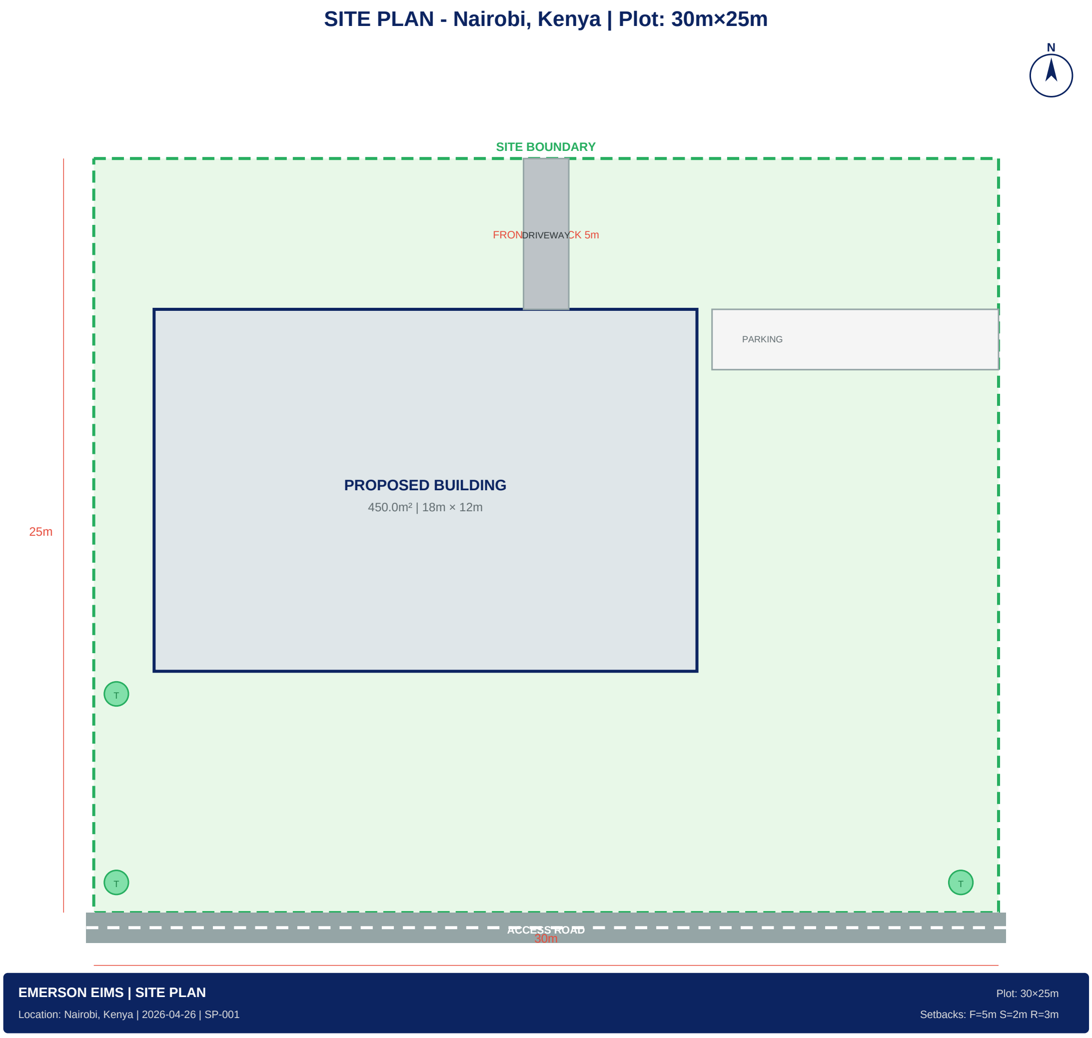
material_cost	60448.52
labor_cost	6800.46
equipment_cost	4835.88
overhead	5379.92
contingency	7746.48
subtotal	85211.26
contractor_profit	12781.69
total_project_cost	97992.94
cost_per_sqm	445.42
labor_multiplier	0.25
region	Kenya
currency	USD

Phase phase_13: Integration Layer

status	complete
export_formats	['IFC2x3', 'FBX', 'DXF', 'PDF', 'XLSX']

2. SITE PLAN

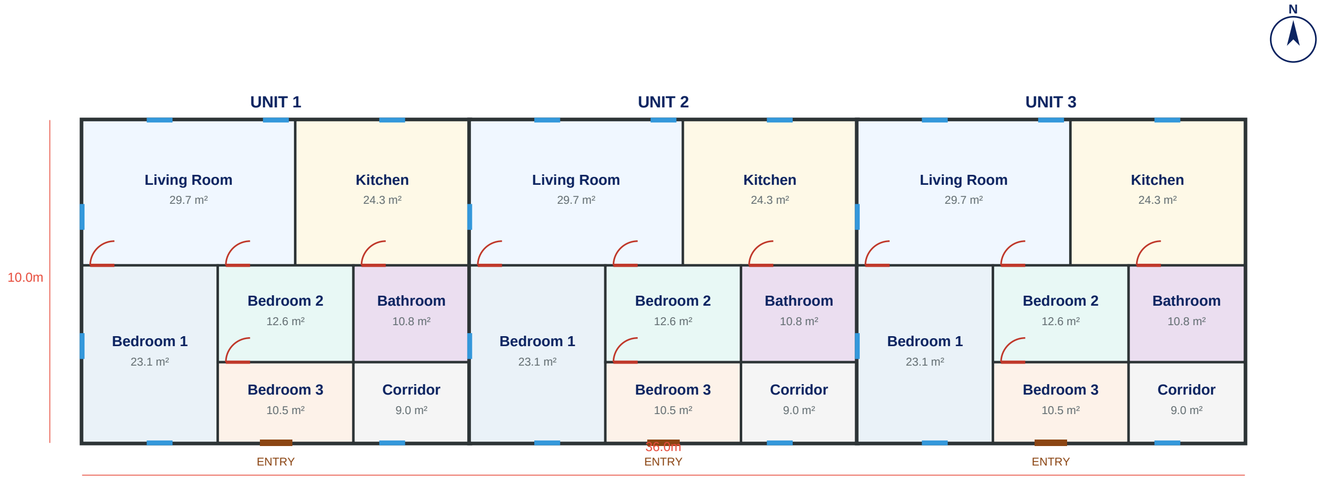
Site layout showing building footprint, access roads, setbacks and landscaping. Area: 450 m²



3. ARCHITECTURAL FLOOR PLAN

Ground floor layout — 3 unit(s), 3 bedrooms, showing room dimensions, doors, windows and circulation.

FLOOR PLAN - 450.0m² | 3 Units | 2 Storey(s)



EMERSON EIMS BUILDING SUITE PRO
Floor Plan | Scale 1:50 | 2026-04-26 | Drawing: FP-001

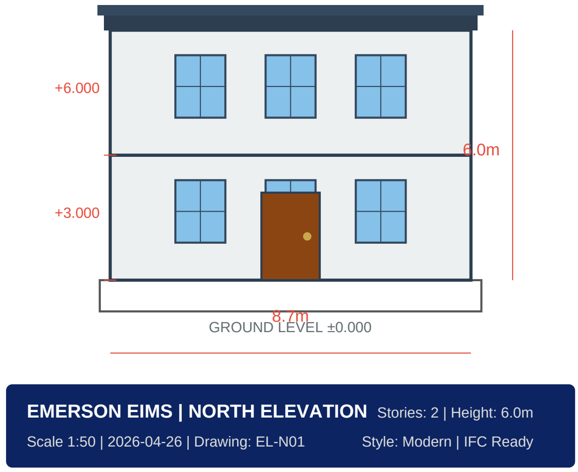
Area: 450.0m² | 3 Units × 3 Bed
IFC Ready | BIM Level 2

4. BUILDING ELEVATIONS

4a. NORTH ELEVATION

Front view showing 2 storey(s), window/door openings, roof profile and floor levels.

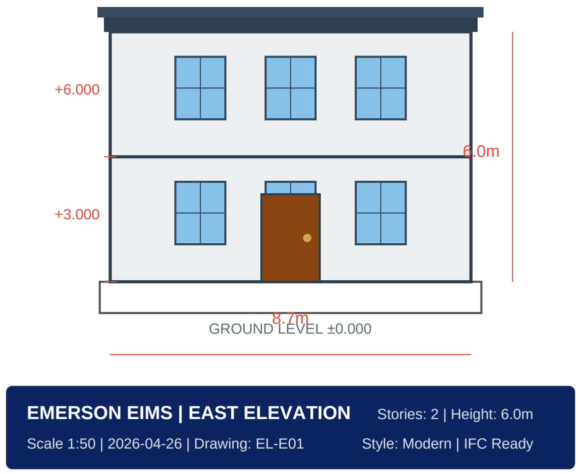
NORTH ELEVATION - 2 Stories | Style: Modern



4b. EAST ELEVATION

Side view showing building depth, roof slope and storey heights.

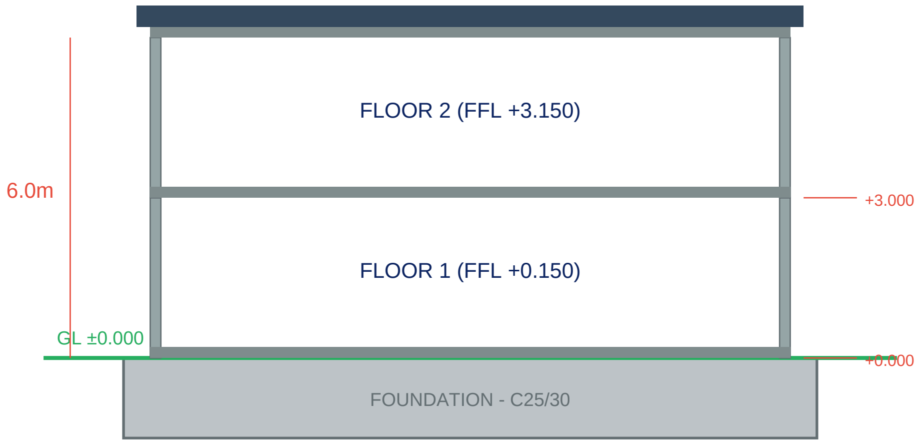
EAST ELEVATION - 2 Stories | Style: Modern



5. BUILDING CROSS SECTION

Vertical cut through building showing floor slabs, storey heights (2 floors), foundation depth, roof structure and structural members.

CROSS SECTION A-A | 2 Stories



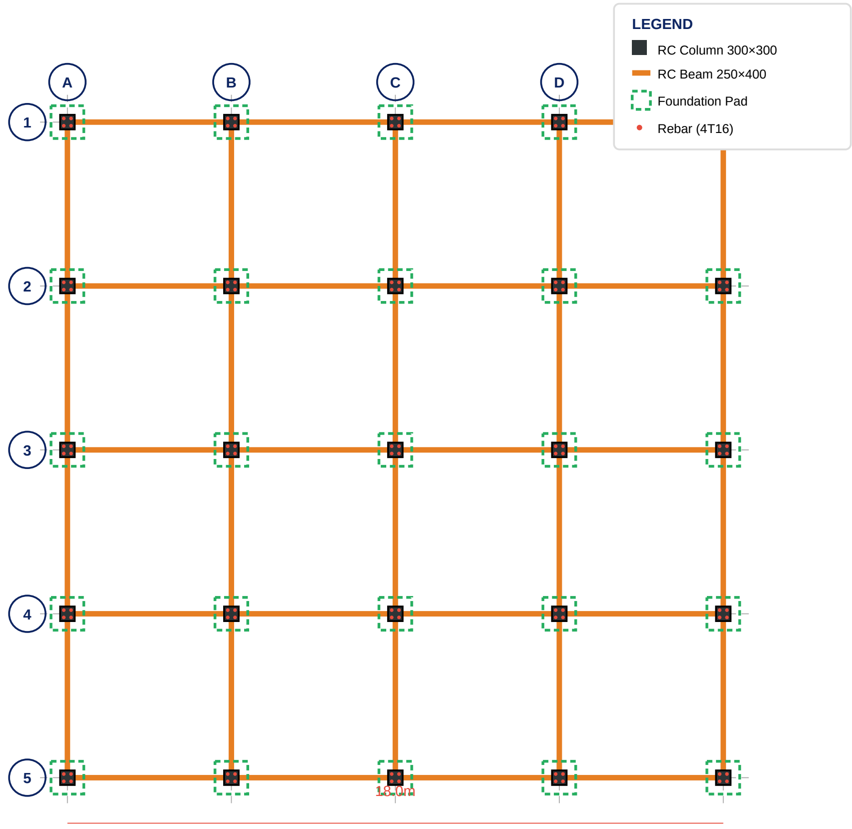
EMERSON EIMS | BUILDING SECTION A-A

2026-04-26 | SEC-001 | Scale 1:50

6. STRUCTURAL LAYOUT

Column grid, beam layout and foundation pads. Foundation type: Strip + Ground Beam. All dimensions in metres. Concrete grade C25/30, Steel Y16.

STRUCTURAL LAYOUT - Strip + Ground Beam Foundation | 2F | Grid 4.5m×4.5m

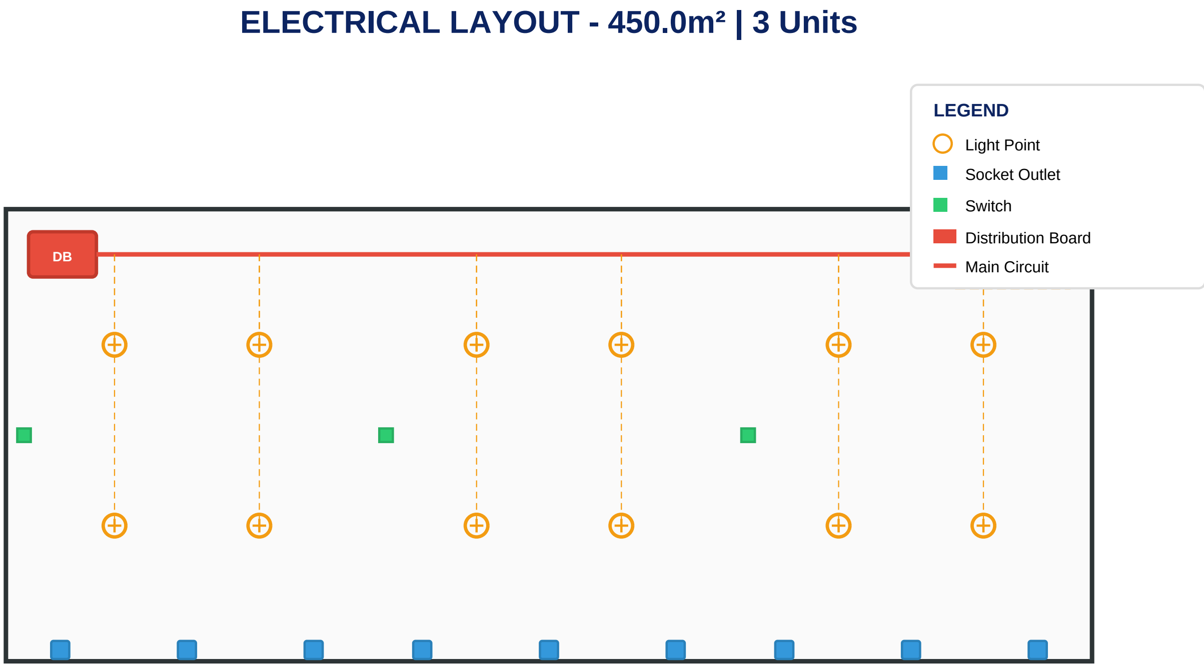


EMERSON EIMS | STRUCTURAL LAYOUT
Grid: 4.5m × 4.5m | Columns: 25 | 2026-04-26 | STR-001

Concrete: C25/30 | Steel: Y16
Foundation: Strip + Ground Beam | FOS ≥ 3.0

7. MEP LAYOUT — ELECTRICAL

Electrical distribution layout showing DB positions, circuit runs, socket/light points and earthing.



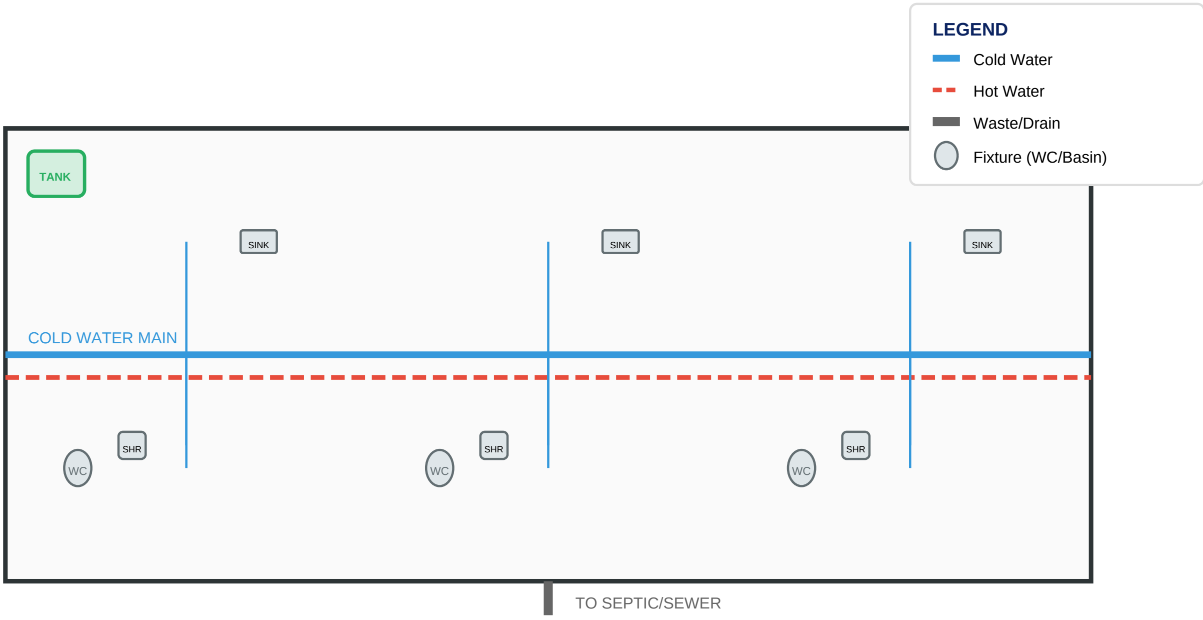
EMERSON EIMS | ELECTRICAL LAYOUT

Scale 1:50 | 2026-04-26 | ELECTRICAL-001

8. MEP LAYOUT — PLUMBING

Plumbing layout showing supply/drain runs, fixture locations, water heater and main shutoff valve.

PLUMBING LAYOUT - 450.0m² | 3 Units



EMERSON EIMS | PLUMBING LAYOUT

Scale 1:50 | 2026-04-26 | PLUMBING-001

9. COST SUMMARY

Cost data not yet generated. Run the analysis first.

This report was generated by EMERSON EIMS Building Suite Pro. All drawings are indicative and must be verified by a licensed engineer before construction.